An Investigation into Paradoxes Between Latent Aptitude and Observed Statistics via Novel Full House Modeling

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This is a hopefully self-contained writeup that contains the foundational material needed to understand what results we need to move this work forward.

Sentences in bold text will denote results that we need investigated.

Sentences in red text will denote attempts that we have made.

Summary

- **Contribution of the paper:**
  - Introduces a statistical framework (Full House Modeling) to fairly compare individuals across contexts (schools, eras, and other domains) where measurement and selection differ.
  - This work formalizes the model that we developed in our paper on comparing baseball players across eras [Yan et al., 2025].
  - Defines and quantifies the *aptitude–achievement paradox*: cases where someone with higher true ability shows lower observed performance across contexts. We define this paradox in terms of representative individuals from two systems drawn from two distinct eligible populations i. For example, this could be government officials elected from two states or baseball players from different seasons.
  - Derives *distribution-free formulas* for paradox probabilities, connects to classic paradoxes (Simpson's, Lord's), and provides modern fairness tools.
- **What we are trying to solve now (generalization of Proposition 3.5):**
  - Extend the proof of the key *inequality condition* that underpins the distribution-free paradox probability formula.
  - Notation: N_i is the total size of population i, and n_i is the number of top individuals from that population selected into the system (so $n_i \leq N_i$).

– The condition is currently proven when $n_1 \leq n_2$ and $N_2 \leq N_1$.

– Goal is to generalize it to the following condition:

$$\frac{n_1}{N_1} \leq \frac{n_2}{N_2},$$

i.e., system 1 selects a smaller fraction of its population than system 2. We have repeatedly verified this result empirically, but cannot yet prove it

– Solving this unlocks the full generality of our paradox probability results for top-selection settings to the best of our knowledge.

What follows is the current version of the paper up to Proposition 3.5. Proofs of results are included for completeness.

Introduction

The following paragraph introduces the intuition behind Full House Modeling. It is adapted from Yan et al. [2025] with some slight changes:

Comparisons of human performance across differing contexts, schools, regions, eras, institutions, and tasks, are central to education, labor economics, psychometrics, medicine, and sports analytics [Chetty et al., 2014, Iezzoni et al., 1997, Bartholomew et al., 2011, Bond and Fox, 2013, Berry et al., 1999, Imbens and Menzel, 2018, Holland and Wainer, 2012, Yan et al., 2025]. A recurring challenge is that an individual in one context may have higher latent aptitude yet lower observed achievement than a counterpart elsewhere. We term this the aptitude-achievement paradox and develop a statistical framework to quantify its probability across contexts. Formally, let individual j in context i have an observed $\mathbb{R}$-valued statistic $Y_{i,j}$ and an underlying latent $\mathbb{R}$-valued aptitude $X_{i,j}$, and let individual n in context m has an observed $\mathbb{R}$-valued statistic $Y_{m,n}$,and an underlying latent $\mathbb{R}$-valued aptitude $X_{m,n}$. The probability of aptitude-achievement paradox is

$$\mathrm{P}\left(X_{i,j} > X_{m,n}, Y_{i,j} < Y_{m,n}\right)$$

The paradox is structurally impossible if a single, strictly increasing, context-invariant measurement map aptitude to achievement and if selection into tasks is ignorable. In practice, however, both measurement heterogeneity (non-invariance, differential item functioning) and non-ignorable selection (who attempts which tasks, faces which opponents, receives which treatment) are pervasive. These features can reverse cross-context rankings even when the latent ranking is unambiguous, echoing classic reversal phenomena such as Simpson's [Simpson, 1951] and Lord's paradoxes [Lord, 1967] while implicating modern concerns about fairness, transportability, and domain shift. What practitioners lack is a principled way to quantify how often such reversals should occur under an explicit model of aptitude, measurement, and selection.

A large class of methods measures how often observed orderings disagree with a target ordering-precisely the phenomenon at the heart of our cross-context paradox. At the

most elementary level, discordance statistics summarize the share of ordered pairs whose observed outcomes align (or misalign) with a reference score or latent ordering. Classic measures include Kendall's τ [Kendall, 1938] and Somers'D [Somers, 1962], which compute differences between the proportions of concordant and discordant pairs and offer adjustments for ties; Somers'D is asymmetric and treats one variable (e.g., a score) as the predictor of the other (e.g., an outcome), making it particularly natural for "does the score rank the outcomes correctly?" questions. These rank-based indices provide assumption-lean, effect-size-like summaries of reversal risk. However, because these indices operate on observed rankings, they neither place individuals on a common latent scale nor disentangle measurement non-invariance or selection, motivating models that make the measurement process explicit.

Traditional latent trait models [Bartholomew et al., 2011] focus on estimating underlying traits from observed categorical data. One of the example of traditional latent trait models, the Rasch model [Bond and Fox, 2013] ensures that comparisons of individuals' abilities are independent of the specific items administered, and the Rasch model expect no differential item functioning (DIF) [Holland and Wainer, 2012], which means an item behaves differently for two (or more) groups of examinees who have the same underlying ability on the target construct. But it does not explicitly leverage the source population structure. Likewise, the bayesian hierarchical modeling [Berry et al., 1999] with the application in comparison of individuals' latent aptitudes typically captures population-level variation through seasonal random effects. In parallel, the causal bootstrap [Imbens and Menzel, 2018] highlights that valid causal inference similarly requires situating the observed sample within large source population to properly account for sampling uncertainty.

Full House Modeling offers a novel approach for comparing individuals across diverse contexts, initially developed through the analysis of baseball players across different eras [Yan et al., 2025]. This analytical framework gained further historical significance [Burgos Jr et al., 2025], which was subsequently explored and presented in a separate study. These analyses are transforming baseball analytics and are increasingly recognized within the broader baseball community. The applicability of Full House Modeling was further extended to investigate the influence of historical figures [Cohen et al., 2025].

In this article, we formalize the Full House Methodology and investigate the probabilities of observing paradoxes between latent aptitude and observed achievement in comparative analysis of individuals across contexts. The aptitude-achievement paradox arises when an individual from one context demonstrates higher latent aptitude but simultaneously lower observed achievement than an individual from another context. We provide explicit probabilities for these paradoxes in two scenarios: first, when observed achievement is independent and identically distributed (i.i.d.), and second, when observed statistics across contexts follow different normal distributions but remain independent. We also explore bounds for these paradoxes in more generalized settings and demonstrate that when observed achievement is normally distributed, paradox probabilities can be reliably estimated using a plug-in method.

Model Setting

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying latent $\mathbb{R}$-valued aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{iid}{\sim} F_X$. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j. We then define an indicator function $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right)$ as a determined and known system inclusion function for population i where a 1 indicates inclusion and a 0 indicates exclusion. We now suppose that an individual j included in system i has an observed $\mathbb{R}$-valued statistic $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$, $n_i \leq N_i$. As we will see later, the observed $Y_{i,j}$ values are one-to-one realizations of $X_{i,j}$, provided that $g_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1$. Therefore $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$.

Let $X_{i,(j)}$ and $Y_{i,(j)}$ denote the jth order statistic for system i. Here, larger numbers will correspond to a notion of "better." Thus $X_{i,(j)} > X_{i,(k)}$ whenever $j > k$, for example. We now form (Y, X) pairs of random variables for each individual included in the system. These pairs are of the form $(Y_{i,(j)}, X_{i,(r_{i,j})})$ where $r_{i,j}$ indexes the aptitude of the jth highest value observed from the system. As an example, let $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$, then the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs take the form $(Y_{i,(j)}, X_{i,(N_i - n_i + j)})$ for $j = 1, \ldots, n_i$.

The key Full House Modeling is the explicit connection between the elements forming the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs. Full House Modeling states this connection as

$$\begin{aligned} Y_{i,(j)} &= F^{-1}_{Y_{i,(j)}} \circ F_{X_{i,(r_{i,j})}}\left(X_{i,(r_{i,j})}\right), \\ X_{i,(r_{i,j})} &= F^{-1}_{X_{i,(r_{i,j})}} \circ F_{Y_{i,(j)}}\left(Y_{i,(j)}\right), \end{aligned} \tag{1}$$

where $1_i(X_{i,(r_{i,j})}, \mathbf{X}_{i,-(r_{i,j})}) = 1$, and both F_{Y_i} and F_X are continuous and invertible. We summarise Full House Modeling in the definition below.

Definition 0.1 (Full House Model)**.** *Consider populations with sizes N_i and system sizes n_i, $N_i \geq n_i$. Let a system inclusion function $1_i(\cdot, \cdot)$ be specified. Let $X_{i,j} \overset{iid}{\sim} F_X$, $j = 1, \ldots, N_i$. Let $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$. Form pairs $(Y_{i,(j)}, X_{i,(r_{i,j})})$, $j = 1, \ldots, n_i$ for all individuals such that $1_i(X_{i,(r_{i,j})}, \boldsymbol{X}_{i,-(r_{i,j})}) = 1$. Then the Full House Model generates $Y_{i,(j)}$ values following (1) where F_{Y_i} and F_X are continuous and invertible.*

Note that Definition 0.1 fixes F_X but allows F_{Y_i} to vary across systems.

Paradoxes Under Identical Distributions

In this section we will provide explicit paradox probability results in a Full House Modeling setting involving two populations with the additional specification that $F_{Y_i} = F_Y$, $i = 1, 2$ in Definition 0.1. The following definition formally defines the notion of paradox used throughout this article.

Definition 0.2. *Consider the Full House Model in Definition 0.1 with two populations denoted by $i = 1, 2$. We say a paradox has occurred when both $X_{1,(r_{1,j})} > X_{2,(r_{2,k})}$ and $Y_{1,(j)} < Y_{2,(k)}$ have occurred. The paradox probability is defined as*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right). \tag{2}$$

As a simple illustrative example of the above, let $r_{i,t} = N_i - n_i + t$ and set $j = n_1$ and $k = n_2$, and further suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then the paradox probability (2) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ when $N_1 \to \infty$. We now introduce a convenient representation which allows the paradox probability to be expressed in terms of the latent aptitudes alone.

Lemma 0.3. *Consider the Full House Model in Definition 0.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Then,*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right), \qquad (3)$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$.

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}), f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{1,2,j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. □

Our first formal result provides conditions that allow for an explicit paradox probability result. Those conditions require the following definition.

Definition 0.4. *Let $\alpha, \beta > 0$. Then the regularized incomplete beta function is defined as*

$$I_{\alpha,\beta}(x) = \frac{1}{B(\alpha,\beta)} \int_0^x u^{\alpha-1}(1-u)^{\beta-1} du,$$

where $B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the beta coefficient.

Note that the regularized incomplete beta function is the cumulative distribution function of a Beta(α, β) distribution.

Theorem 0.5. *Consider the Full House Model in Definition 0.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Denote by I the regularized incomplete beta function. Further assume that*

$$I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \geq 0, \qquad \text{for all } t \in (0,1). \qquad (4)$$

Under these assumptions,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&= \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2-n_1-p+k+1)}{B(k, n_2-k+1)} - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2-N_1-p+r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}$$

Remarks:

1. The result of Theorem 0.5 is distribution free. It only depends on the sample sizes n_i, N_i, $i = 1, 2$, the system inclusion function, and the indexes of the order statistics under comparison. The distribution on Y is eliminated by Lemma 0.3. To see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F_Y^{-1} \circ U_{(j)}$ where $U_{(j)} \sim \text{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = I_{j,n_1-j+1} \circ F_Y(y)$. With this in mind, from Lemma 0.3 we have that

$$\begin{aligned} h_{1,2,j,k}(x) &= F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(x) \\ &= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ \left(F_Y \circ F_Y^{-1}\right) \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x) \\ &= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x). \end{aligned}$$

   The above is the upper-bound of an integral with respect to Xs in the proof of Theorem 0.5. The influence of the distributional specification on Y does not influence the calculation of the paradox probability result in Theorem 0.5.

2. The distribution on X is eliminated when calculating the paradox probability (2) in its Lemma 0.3 representation, $\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right)$. These calculations involve double integrals with respect to densities for $F_{X_{1,(r_{1,j})}}$ and $F_{X_{2,(r_{2,k})}}$. One integral evaluation and a reparameterization remove the specification of F_X from the probability calculation as seen in the proof of Theorem 0.5.

3. The steps mentioned in the last sentence of the previous remark allow the probability calculation for $\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right)$ to simplify to

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt.$$

   The technical condition (4) ensures that this integral defines a non-negative probability in strong terms. We demonstrate with analytic and numeric arguments when (4) is satisfied and the results of Theorem 0.5 therefore hold.

Proof. First, Lemma 0.3 allows us to write

$$\text{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. We can rewrite $h_{1,2,j,k}(\cdot)$ into a form that will allow for specific computation involving Beta distributions. o see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F_Y^{-1} \circ U_{(j)}$ where $U_{(j)} \sim \text{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = g_{n,j} \circ F_Y(y)$, where we write $g_{n,j}(x) = I_{j,n-j+1}(x)$ for ease of notation. This expression of distribution functions allows us to write

$$\begin{aligned} &\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right) \\ &= \int_{-\infty}^{+\infty} \left\{ \int_x^{F_X^{-1} \circ g^{-1}_{N_1,r_{1,j}} \circ g_{n_1,j} \circ g^{-1}_{n_2,k} \circ g_{N_2,r_{2,k}} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \end{aligned}$$

$$
\begin{aligned}
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1}\circ g_{N_1,r_{1,j}}^{-1}\circ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x)} d\ g_{N_1,r_{1,j}}(F_X(y)) \right\}\ f_{X_{2,(r_{2,k})}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x) - g_{N_1,r_{1,j}}\circ F_X(x) \right\}^{+}\ f_{X_{2,(r_{2,k})}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x) - g_{N_1,r_{1,j}}\circ F_X(x) \right\}^{+}\ d\ g_{N_2,r_{2,k}}(F_X(x)) \\
&= \int_{0}^{1} \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^{+}\ dt.
\end{aligned}
$$

Condition (4) states that

$$
\int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^{+}\ dt = \int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}\ dt.
$$

Thus,

$$
\begin{aligned}
&\int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^{+}\ dt \\
&\quad = \int_0^1 g_{n_1,j}(x)\ dg_{n_2,k}(x) - \int_0^1 g_{N_1,r_{1,j}}(x)\ dg_{N_2,r_{2,k}}(x) \\
&\quad = \int_0^1 I_x(j, n_1-j+1)\frac{x^{k-1}(1-x)^{n_2-k}}{B(k,n_2-k+1)}\ dx \\
&\qquad - \int_0^1 I_x(r_{1,j}, N_1-r_{1,j}+1)\frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(k,N_2-k+1)}\ dx \\
&\quad = \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p(1-x)^{n_1-p}\frac{x^{k-1}(1-x)^{n_2-k}}{B(k,n_2-k+1)}\ dx \\
&\qquad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p(1-x)^{N_1-p}\frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k},N_2-r_{2,k}+1)}\ dx \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2+n_1-p-k+1)}{B(j,n_2-k+1)} \\
&\qquad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k},N_2-r_{2,k}+1)}.
\end{aligned}
$$

□

Often it is of interest to compare the individuals with the highest aptitudes in their respective populations. Below we state the paradox probability in such setting using a Full House Model with two populations with the specification that $F_{Y_1} = F_Y$ for $i = 1, 2$.

Corollary 0.5.1. *Suppose that the conditions of Theorem 0.5 hold. Suppose that* $1_1(X_{1,(N_1)}, \boldsymbol{X}_{1,-(N_1)}) = 1_2(X_{2,(N_2)}, \boldsymbol{X}_{2,-(N_2)}) = 1$. *Then,*

$$
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \le \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.
$$

A similar result holds for the individuals with the lowest latent aptitudes in the population.

Corollary 0.5.2. *Suppose that the conditions of Theorem 0.5 hold. Suppose that* $1_1(X_{1,(1)}, \boldsymbol{X}_{1,-(1)}) = 1_2(X_{2,(1)}, \boldsymbol{X}_{2,-(1)}) = 1$. *Then,*

$$\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \begin{cases} \frac{N_2}{N_1+N_2} - \frac{n_2}{n_1+n_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.$$

We now demonstrate a setting in which technical condition (4) holds. This setting is based on a configuration of n_j, N_j, $j = 1, 2$ under the assumption that the systems are populated with individuals from the source populations who have the highest aptitudes.

Proposition 0.6. *Consider the Full House Model in Definition 0.1 with two populations, where* $F_{Y_i} = F_Y$ *for* $i = 1, 2$, *and let the system inclusion function be* $1_i(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$. *Denote by* I *the regularized incomplete beta function. Suppose that* $\frac{n_1}{n_2} \leq 1$ *and* $\frac{N_2}{N_1} \leq 1$, *and set* $k = j$. *Under these assumptions, we conclude that technical condition (4) holds and the conclusion of Theorem 0.5 applies.*

Proof. Specification of $1_i(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$ implies that $r_{i,j} = N_i + 1 - j$.

When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $x \in (0, 1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased [Askitis, 2021], or n_2 or N_1 are decreased. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned} & I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\ & \quad \implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\ & \quad \implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t. \end{aligned}$$

Technical condition (4) holds and the conclusion of Theorem 0.5 applies. □

We think that Proposition 3.5 can be generalized to $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$ with a different argument.

We tried the following in an attempt to generalize:

Let

$$f_{n_1,n_2,N_1,N_2,j}(x) = I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x).$$

Applications of the chain rule, the inverse function theorem [Rudin et al., 1964] on Page 221, and this property of regularized incomplete beta functions

$$I'_{\alpha,\beta}(I^{-1}_{\alpha,\beta}(x)) = \frac{I^{-1}_{\alpha,\beta}(x)^{\alpha-1}(1 - I^{-1}_{\alpha,\beta}(x))^{\beta-1}}{B(\alpha, \beta)},$$

allow us to write

$$
\begin{aligned}
f'_{n_1,n_2,N_1,N_2,j}(x) &= \frac{I'_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x)}{I'_{n_2-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x)} - \frac{I'_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x)}{I'_{N_2-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x)} \\
&= \frac{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-j-1}\left(1-I^{-1}_{n_2-j,j+1}(x)\right)^{j}/B(n_1-j,j+1)}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_2-j-1}\left(1-I^{-1}_{n_2-j,j+1}(x)\right)^{j}/B(n_2-j,j+1)} \\
&\quad - \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-j-1}\left(1-I^{-1}_{N_2-j,j+1}(x)\right)^{j}/B(N_1-j,j+1)}{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_2-j-1}\left(1-I^{-1}_{N_2-j,j+1}(x)\right)^{j}/B(N_2-j,j+1)} \\
&= \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)}\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2} - \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)}\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2} \\
&= \left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}\frac{B(N_2-j,j+1)}{B(N_1-j,j+1)} \\
&\quad \times \left[\frac{B(n_2-j,j+1)/B(n_1-j,j+1)}{B(N_2-j,j+1)/B(N_1-j,j+1)} - \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}\right]
\end{aligned}
$$

From here we see that $f_{n_1,n_2,N_1,N_2,j}(0) = f_{n_1,n_2,N_1,N_2,j}(1) = 0$, and when $x = 1$, we have

$$
\begin{aligned}
f'_{n_1,n_2,N_1,N_2,j}(1) &= \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)} - \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)} \\
&= \frac{(n_1-j)\ldots(n_1)}{(n_2-j)\ldots(n_2)} - \frac{(N_1-j)\ldots(N_1)}{(N_2-j)\ldots(N_2)} \\
&\leq 0,
\end{aligned}
$$

from the assumption that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$. If we can prove the $\frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}$ is a monotonic decreasing function for $x \in (0,1)$ and ranges from 0 to 1, then we can conclude that $f_{n_1,n_2,N_1,N_2,j}(x) \geq 0$ as desired.

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